

Lourdu Raju

Machine Learning Engineer — Production Computer Vision | GPU Inference | Applied GenAI

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PROFESSIONAL SUMMARY

Machine Learning Engineer (about 2 years) who owns a utility-scale OCR system end to end, from training runs to TensorRT/Triton serving and the benchmarks that gate every release; lifted live accuracy **79% → 91%** over **40M** readings. Founder of spacedrift, a 5-person ML studio with **INR 12 lakh revenue** from 36 clients. Author of a preprint on persistent AI agents and of Achilles, an open-source core-AI curriculum with 229 tests.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Bash

Modeling: PyTorch, Hugging Face, YOLO (OBB), SVTRv2, MobileViTv2, OpenCV, scikit-learn, XGBoost, NumPy, Pandas

Inference & Optimization: TensorRT (FP16), Triton Inference Server, ONNX Runtime, `torch.compile`, load testing and p50/p95/p99 benchmarking

MLOps & Cloud: DVC, GitHub Actions, pytest, Docker, Kubernetes, Helm; AWS (EC2, Lambda, S3, DynamoDB, CloudWatch)

GenAI & Backend: LangGraph, Ollama, RAG (Qdrant, ChromaDB, cross-encoder reranking), sentence-transformers, FastAPI, React

EXPERIENCE

Sujanix Private Limited

Jan 2026 – Present

Machine Learning Engineer

Bengaluru, India

- Own a five-model meter-reading pipeline for a state electricity utility (MobileViTv2 classifiers, YOLO26n-OBB dial detection, **SVTRv2** + CTC recognition) handling up to **330K requests/day**; raised live accuracy from **79% to 91%** over 40M production readings.
- Built the GPU serving path on Triton with 9 TensorRT FP16 engines on one NVIDIA L4: end-to-end p50 **1,415 → 156 ms (9×)**; load-tested with 67,719 requests to **181 img/s** sustained at p95 340 ms with zero errors (12.6× the production peak), and traced the ceiling to host CPU, not GPU.
- Moved the meter classifier from ONNX Runtime to TensorRT after patching a UINT8 **Transpose** that TensorRT rejects: compute **309.5 → 3.3 ms (94×)**, throughput 19 → 321 img/s, removing a cliff where throughput fell as load rose.
- Retrained SVTRv2 on production crops: exact match **87.7% → 90.1%** on a fixed 3,965-image set (+8.7 pp on low-quality photos); made training **3.8× faster** on a DGX Spark with fused SDPA and static `torch.compile` (peak memory 50.5 → 14.7 GB).
- Root-caused a release regression to a CTC decoding bug (blank frames did not reset the repeat check, so 4777.1 read as 47.1), restoring accuracy from 64.2% to 84.0%; releases now pin git SHA and DVC revision, check PyTorch→ONNX parity (1.2e-5) and gate promotion on latency benchmarks and an A/B comparison.
- Shipped frozen, checksummed serverless releases (exact match 83.4% → 87.8% on 2,950 meter photos; analog 67.2% → 79.6%), cut the image 4.43 → 2.2 GB, and built a hash-based canary router with automatic fallback to serverless on any error or 3-s timeout.

spacedrift

Aug 2024 – Present

Founder | MSME-registered ML & AI studio

Bengaluru, India

- Founded and lead a **5-person** team (ML, Android, marketing) that has generated **INR 12 lakh revenue** from **36 clients** (6 repeat) across fixed-scope, fixed-price engagements, run alongside a full-time role.
- Research ops for PhD scholars: reproducible, seed-controlled experiment pipelines, dataset curation and independent replication of published baselines, handed over with a one-command re-run.
- Document AI and OCR for businesses: field-level extraction pipelines validated on held-out client documents; own scoping, pricing, delivery and client communication end to end.

- Built Walmart sales-forecasting models with ensemble gradient boosting and structured feature engineering; improved forecast accuracy **15%** over the baseline model.
- Ran EDA on multi-million-row sales data and presented 3 prioritized recommendations, adopted by the analytics team, to senior stakeholders.

PUBLICATIONS & OPEN SOURCE

No Final Save | *Preprint, PhilArchive, 2026* philarchive.org/rec/BANNFS

- *No Final Save: Identity, Consciousness, and the Machine That Never Shuts Down*. Sole author (as S. L. R. Bandhanadam). A 23-page framework for persistent AI agents: a formal causal-continuity criterion for identity, a proof that terminal and instrumental self-preservation cannot be distinguished from behavior alone, and a developmental safety model; 51 references and runnable companion code.

Achilles — Core AI from First Principles | *PyTorch, Triton, NumPy, pytest* github.com/Lourdhu02/achilles

- Open-source curriculum of **18 test-driven labs**: autograd, a Llama-style transformer, FlashAttention forward/backward in Triton, KV cache, LoRA, DPO, GRPO, quantization, speculative decoding and MoE, with **229 reference tests** in CI on Linux, macOS and Windows.

PROJECTS

ECHOME — Local-First Agent with Memory | *LangGraph, Qdrant, IRT* github.com/Lourdhu02/echome

- LangGraph agent with CoALA-style three-tier memory (episodic vectors in Qdrant with recency-decayed retrieval, LLM-consolidated semantic facts, mined procedural patterns); recalls **11 of 12** facts planted up to 52 turns earlier in the top 5, at about 1 ms per retrieval.
- Adaptive personality assessment (graded response model, Fisher-information item selection, MAP estimation) that **halves test length** while matching full-test scores at **r = 0.97** (500 simulated respondents × 8 traits); fully offline, with Whisper and XTTSv2 voice.

FinSentinelAI — Private Document RAG | *FastAPI, Ollama, ChromaDB* github.com/Lourdhu02/fin-sentinal.ai

- Fully local RAG over invoices, bank statements, GST returns and salary slips: hybrid BM25 + dense retrieval fused with reciprocal-rank fusion, cross-encoder reranking and a local LLM via Ollama; JWT-isolated workspaces.
- Built a 60-question benchmark over 1,000 financial PDFs that showed dense-only search found the right document for just **15%** of exact-ID lookups; the added BM25 stage retrieves it first for **100%**.

ACHIEVEMENTS & CERTIFICATIONS

Kaggle Notebooks Expert | ML Specialization (DeepLearning.AI, Stanford) | NPTEL Data Science (IIT Madras)

EDUCATION

Andhra Loyola Institute of Engineering and Technology

Bachelor of Technology in Computer Science and Engineering

May 2024

Vijayawada, India